

Rajendra Bist

Full-Stack Web Developer & Backend Engineer

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SUMMARY

Full-stack developer trained at Vcare Technical Institute with hands-on experience building real-time systems, REST APIs, and AI-integrated products. Shipped a WebSocket-based messaging application, a Gemini-powered Socratic AI learning assistant with streaming LLM responses, and a personal portfolio with a live AI chat agent backed by MongoDB. Currently deepening backend expertise in Node.js architecture, database schema design, authentication systems, and LLM integration. Open to backend-heavy full-stack roles and AI-integrated product work.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript, Python (basic), HTML5, CSS3
Frontend:	React 18, Next.js 15 (App Router), Tailwind CSS, Framer Motion, Responsive Design
Backend:	Node.js, Express.js, REST API Design, WebSockets (Socket.io), JWT Authentication
Databases:	MongoDB (primary), Mongoose ODM, PostgreSQL (learning), Database Schema Design
AI / LLM:	Google Gemini API, Streaming Interfaces, Prompt Engineering, Vercel AI SDK (learning)
Tools:	Git, GitHub, Vercel, Railway, Docker (learning), Linux CLI

PROJECTS

SocraticAI — AI-Powered Socratic Learning Assistant

Next.js 15 · Node.js · Google Gemini API · MongoDB · Tailwind CSS | [GitHub](#) | Live: socratic-ai-rho.vercel.app

- Built and deployed a chat-first AI learning assistant applying the Socratic method — guiding users to deeper understanding through structured counter-questions rather than direct answers.
- Integrated Google Gemini API with custom system prompt engineering to enforce Socratic dialogue; implemented streaming responses for real-time token-by-token output mirroring production AI chat UX.
- Full-stack architecture: Next.js 15 App Router, Node.js backend, MongoDB Atlas for conversation logging with TTL auto-expiry, in-memory rate limiting, and Vercel serverless deployment.

WhatsApp Clone — Real-Time Messaging Application

React · Node.js · Express.js · Socket.io · Tailwind CSS | Live: whatsapp-clone-client-liart.vercel.app

- Built a real-time messaging system using Socket.io with room-based message routing and multi-user session management, demonstrating bidirectional WebSocket event architecture.
- Custom event protocol (join-room, send-message, disconnect) handling concurrent users; separated frontend (Vercel) and backend (Railway) deployments with proper CORS configuration.

Developer Portfolio with AI Chat Agent

Next.js 15 · Gemini API · MongoDB · Framer Motion | Live: rajendra-bist.vercel.app

- Production portfolio with a floating AI chat agent (Gemini 1.5 Flash) that streams context-aware responses about skills and projects using a custom system prompt and Web API ReadableStream.
- Dark/light theme toggle, Framer Motion scroll-reveal animations, contact form with MongoDB persistence — fully responsive across mobile and desktop.

Todo Application

Next.js · React · Tailwind CSS | Live: todo-app-azure-five-70.vercel.app

- Task management app demonstrating React component architecture, CRUD state management, and optimistic UI updates with localStorage persistence.

EDUCATION & TRAINING

Full-Stack Web Development	Vcare Technical Institute · <i>Next.js, Node.js, React, MongoDB, Express.js, REST APIs, PostgreSQL, System Design</i>
+2 Science	KMC — Kailali Multiple Campus
Secondary Education (SEE)	Shree Phulwari Secondary School

ADDITIONAL

- **Currently Building:** Expanding SocraticAI with advanced dialogue flows, user progress tracking, and topic-specific learning paths
- **Backend Focus:** Node.js architecture, MongoDB schema design, JWT auth systems, system design fundamentals, microservice patterns
- **Location:** Nepal — available for remote roles worldwide; open to discussing onsite/hybrid opportunities
- **Languages:** Nepali (Native), Hindi (Fluent), English (Professional)